MIL-P-43327B

18 March 1977

SUPERSEDING
MIL-P-43327A
11 September 1969

MILITARY SPECIFICATION

PAPER, BOOK, WATER RESISTANT (TEXT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of water resistant book paper (JCP A-220) and reflects item requirements parallel to the specifications of the Joint Committee on Printing.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-P-25 - Printing and Stationery Paper, Packaging and Packing of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Government Paper Specification Standards

Published by the Joint Committee on Printing, Congress of the United States

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

T 401 os-74 - Fiber Analysis of Paper and Paperboard.
T 402 os-70 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets, and Related Products.
T 403 os-76 - Bursting Strength of Paper.
T 409 os-75 - Machine Direction of Paper and Paperboard.
T 410 os-68 - Weight Per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 411 os-76 - Thickness (Caliper) of Paper and Paperboard.
T 412 su-69 - Moisture in Paper.
T 425 os-75 - Opacity of Paper (15°/Diffuse Illuminant A).
T 433 os-74 - Water Resistance of Sized Paper and Paperboard (Dry Indicator Method).
T 435 su-68 - Hydrogen Ion Concentration (pH) of Paper Extracts-Hot Extraction Method.
T 452 m-58 - Brightness of Paper and Paperboard.
T 456 os-68 - Wet Tensile Breaking Strength of Paper and Paperboard.
T 459 os-75 - Surface Strength of Paper (Wax Pick Test).
T 479 su-71 - Smoothness of Paper (Bekk Method).

(Copies of TAPPI Testing Methods may be obtained from the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Standard sample. The color, finish, formation and cleanliness of the paper must conform to the standard sample as adopted by the Joint Committee on Printing as furnished by the contracting officer (see 6.3).

3.2 Paper, book. The book paper shall conform to all the requirements as set forth in the Government Paper Specification Standards for JCP A-220, Printing Paper, Water Resistant (Text) Book.

* 3.3 Workmanship. The sheets shall be trimmed clean; shall not curl; shall not have any tears, cuts, wrinkles, folds, chafed or scuff marks. In addition, there shall be no presence of lint, fuzz, or any particles that will pick, lift, fluff or pile on the press blanket.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, a lot shall consist of all paper of the same type, size and weight.

* 4.2.1 Component and material inspection. The contractor is responsible for insuring that all materials and components used were manufactured and inspected in accordance with the requirements of this specification and referenced specifications.

4.2.2 Inspection of the end item.

4.2.2.1 Examination of the end item. Examination of the end item shall be made for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth in table I.

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* TABLE I. Sampling procedure

Examination paragraph	End item	Lot size in units of	Sample unit	Inspection level	AQL	Defects in 100 units (DHU) of
4.2.2.1.1	8 by 10-1/2 in. (203 by 267 mm) sheet	Pounds (kg)	One 8 by 10-1/2 in. (203 by 267 mm) sheet	S-4	4.0	8 by 10-1/2 in. (203 by 267 mm) sheet
	Larger than 8 by 10-1/2 in. (203 by 267 mm) sheet	Pounds (kg)	One 8 by 10-1/2 in. (203 by 267 mm) sheet	S-2 and S-4 1/	4.0	8 by 10-1/2 in. (203 by 267 mm) sheets
4.2.2.1.2	All size sheets	Reams	Full size Sheet	S-4	4.0	Sheets
4.2.2.1.3	All size sheets	Reams	Full size sheet	S-2	2.5	Sheets
4.2.2.1.4	All size sheets	Reams	Ream or quantity between markers (when skid packed)	S-2	---	---
4.2.2.1.5	All size sheets	Shipping containers	Shipping container	S-2	2.5	Shipping containers
4.2.2.1.6	All size sheets	Shipping containers	Shipping container	S-2	2.5	Shipping containers

1/ For sheets larger than 8 by 10-1/2 inches (203 by 267 mm), full size sheets shall be selected at the S-2 level and cut into as many sheets as possible, 8 by 10-1/2 inches (203 by 267 mm) and randomly arranged for selection of the examination samples at the S-4 level.

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* 4.2.2.1.1 Examination of the end item for color, finish, formation and cleanliness. Only one sheet, 8 by 10-1/2 inches (203 by 267 mm) or larger, selected in accordance with table I shall be taken from a package. Both sides of the sample unit shall be examined.

Examine	Defects
Appearance	Cleanliness, finish or formation (as applicable) does not compare favorably with standard sample.
	Color not uniform on both sides; does not compare favorably with the standard sample.
	Presence of holes, dirt or oil spots, lumps, foreign or undigested matter, shives, scales, shiners, scabs, slivers, slime spots or mottle larger than or more than in standard sample.

4.2.2.1.2 Examination of the end item for workmanship. No more than one sheet, selected in accordance with table I, shall be taken from a ream.

Examine	Defects
Workmanship	Sheets not trimmed clean; ragged, torn, uneven or crushed edges.
	Paper curls, does not lie flat, i.e., not flat after reasonable flexing.
	Any tear, cut, wrinkle, fold, chafed or scuffed mark.
	Presence of lint, fuzz, or any particles which will pick, lift, fluff or pile on the press blanket.

* 4.2.2.1.3 Examination of the end item for dimensional defects. No more than one sheet selected in accordance with table I, shall be taken from a ream.

Examine	Defects
Sheet size	Less than specified in either direction.
Sheets 8-1/2 by 14 inches (216 by 356 mm) or less	Oversize by more than 1/16 inch (2 mm) in either dimension from size specified.

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Examine	Defects
Sheets over 8-1/2 by 14 inches (216 by 356 mm)	Oversize by more than 1/8 inch (3 mm) in either dimension from size specified.
Squareness	Difference between the length of two diagonals exceeds 1/32 inch (1 mm) for sheets 32 inches (813 mm) or less in length, or exceeds 1/16 inch (2 mm) for sheets over 32 inches (813 mm) length.

4.2.2.1.4 Examination of the end item for count. The average count of sheets per package or quantity between markers, when paper is skid packed, shall be not less than the specified quantity.

4.2.2.1.5 Examination of the end item for color uniformity. Color not uniform within the unit package and throughout the skid or shipping container.

* 4.2.2.1.6 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing and marking requirements comply with PPP-P-25.

* 4.2.3 Testing of the end item. All tests shall be conducted under standard conditions specified in Method T 402 os-70 of TAPPI. The test sample unit shall consist of a sufficient number of sheets to provide not less than 1000 square inches (0.645 m^2) of material. Not more than one sample unit shall be drawn from any one package. The lot size for the purpose of determining the sample size shall be expressed in units of reams or packages of sheets of one type and size. The sample size shall be selected in accordance with the sampling plan below. Test results shall include all values upon which the results are based and there shall be no failure to meet the sample unit or lot average requirements.

Lot size (number of reams or packages of sheets)	Sample size (number test units)
0-50	2
51-500	3
501-35,000	5
35,000 and over	8

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INSTRUCTIONS FOR TESTING

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations XXXXXXX Per sample unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Conditioning for testing		T402os-70			All tests				
Bursting strength, minimum, average									
Dry	3.2	T403os-76		X	20, 10 on each side		Point		
Wet	3.2	4.3.1		X	20, 10 on each side		Point		
Grain direction	3.2	T409os-75	X		1	X	---		
Thickness, average	3.2	T411os-76		X	20		0.0001 in. (0.001 mm)		
Moisture content									
As received	3.2	4.3.2		X	1		0.1 percent		
After conditioning	3.2	4.3.2		X	1		0.1 percent		
Fiber content	3.2	T401os-74 1/		2/	2 on composite		5 percent		
Basis weight, average	3.2	T410os-68		X	1 (average of 1000 sq. in.) (6452 cm^2)		0.1 pound (45.4 g)		
Acidity, pH value	3.2	T435su-68 1/		2/	2 (composite)		0.1 pH unit		
Water resistance, minimum both sides	3.2	T433os-74		X	5 each side		Second		

AMXRE Form 598
8 February 1965

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INSTRUCTIONS FOR TESTING

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations ~~EXAMINE~~ Per sample unit	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver		Pass or Fail	Numerically to Nearest		
Pick resistance, minimum, average									
One side	3.2	T459os-75		X	5		Wax No.		
Other side	3.2	T459os-75		X	5		Wax No.		
Brightness, minimum, average	3.2	T452m-58		X	6		percent		
Finish smoothness, average									
One side	3.2	T479su-71		X	5		1 second		
Other side	3.2	T479su-71		X	5		1 second		
Opacity, minimum, average	3.2	T425os-75		X	10		0.5 percent		
Blocking	3.2	4.3.3	X		1	X			

1/ Composite shall consist of approximately equal pieces of paper selected from each unit.

2/ Requirement is applicable to the composite average.

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MIL-P-43278

8

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4.3 Testing procedures.

4.3.1 Wet burst strength.

4.3.1.1 Specimen preparation. Prior to the wet bursting strength test the test specimens shall be heated for 15 minutes at 105°C followed immediately by soaking for one hour in water at a temperature of 23° ± 2°C. After saturation remove the specimens from the water and blot in accordance with TAPPI Standard T 456 os-68.

* 4.3.1.2 Procedure. After soaking and blotting as described in 4.3.1.1 the specimens for wet burst testing shall be tested in accordance with procedure of TAPPI test T 403 os-76.

4.3.2 Moisture content.

* 4.3.2.1 As received. The moisture content of the paper as received shall be determined in accordance with TAPPI Method T 412 su-69. Weigh the sample in the container, then remove and condition the sample to equilibrium in accordance with TAPPI Method T 402 os-70.

* 4.3.2.2 After conditioning. After conditioning, return the sample to the same airtight container while still in the conditioned room, reweigh, and determine the moisture content according to TAPPI Method T 412 su-69, reporting both the skid moisture content and the equilibrium moisture content.

4.3.3 Blocking. This method is applicable to printing and stationery papers, to determine whether they stick together after being wet.

* 4.3.3.1 Test specimens. A rectangular test specimen 45 by 60 mm in dimensions shall be cut from each of 12 sheets, selected so as to be representative of the lot. If more convenient, rectangles 1-1/2 by 2-1/2 inches (38 by 64 mm) may be used. Twelve (12) such specimens shall be used for test.

4.3.3.2 Equipment.

* 4.3.3.2.1 Base plate. The base plate shall be a 4 by 4 inch (100 by 100 mm) square of plate glass or other flat material which is corrosion-resistant under the conditions of the test.

* 4.3.3.2.2 Top plate. The top plate shall be a flat piece of material like the base plate, 45 by 60 mm in dimensions. If 1-1/2 by 2-1/2 inch (38 by 64 mm) test specimens are used, the top plate shall be 1-1/2 by 2-1/2 inches (38 by 64 mm).

* 4.3.3.2.3 Weight. A weight shall be used of such size that the total mass of the weight plus the top plate shall be equivalent to 1 pound per square inch (6.89 kPa) on the specimens. A pressure variation not exceeding plus or minus 5 percent of the specified pressure will be permitted.

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4.3.3.3 Procedure. Soak the 12 specimens in water at 25° ± 5°C for 15 minutes, taking care that the water has free access to both sides of each specimen. Then remove the specimens from the water, shaking each one gently to remove excess water from the surfaces. Stack the specimens evenly on the base plate, assemble the top plate and weight on the specimens, and allow the assembly to dry. Remove the weight and top plate and carefully examine each specimen for blocking (any adherence to adjacent specimens).

4.3.3.4 Report. Report the number of specimens showing blocking.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The water resistant book paper shall be preserved-packaged in accordance with the applicable level A requirements for printing papers as specified in PPP-P-25 and as indicated in table III of this specification.

5.1.2 Commercial. The water resistant book paper shall be preserved-packaged in accordance with the applicable commercial requirements for printing papers as specified in PPP-P-25.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A packing. The water resistant book paper, preserved-packaged as specified in 5.1, shall be packed in accordance with the applicable level A requirements for printing papers as specified in PPP-P-25 and as indicated in table III of this specification.

5.2.2 Level B packing. The water resistant book paper, preserved-packaged as specified in 5.1, shall be packed in accordance with the applicable level B requirements for printing papers as specified in PPP-P-25 and as indicated in tabel III of this specification.

5.2.3 Commercial packing. The water resistant book paper, preserved-packaged as specified in 5.1, shall be packed in accordance with the applicable level C requirements printing papers as specified in PPP-P-25.

5.3 Marking. In addition to any special marking required by the contract, interior packages and shipping containers shall be marked in accordance with the applicable marking requirements of PPP-P-25.

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* TABLE III. Packaging and packing of water resistant book paper

Ream						Shipping container			
Nomenclature	Size in. and (mm)	Weight lbs. and (kg)	Approx. thickness in. and (mm)	Wrapped	Ream marked w/marker placed lengthwise	Approx. inside length and width (inside of liner in. (mm)	Quantity	Approx. net wt. lbs. (kg)	Type
Paper water resistant book	21x16 (533x406)	21.1 (9.57)	2.1 (53)	No	Yes	21 x 32 (533x813)	8 reams	169 (76.6)	Fiberboard
	50x38 (1270x965)	120 (54.4)	2.1 (53)	No	Yes	50 x 38 (1270x965)	2 reams	240 (108.9)	Fiberboard

6. NOTES

* 6.1 Intended use. Water resistant (text) book paper is intended to be used for multicolor face and back letterpress and offset printing of books, pamphlets, manuals, etc., requiring water-resistant properties. Printed matter may include text, line illustrations and halftones up to 150-line screen, not suitable for halftone production by letterpress printing.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Grain direction (see 3.2).
(c) Pressroom conditions (see 3.2).
(d) Selection of the applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 Standard comparison samples may be obtained upon application to the procuring activity issuing the invitations for bids (see 3.1).

* 6.4 Changes from previous issue. The outside margins of this specification are marked with an asterisk (*) to indicate where changes (additions, corrections, modifications, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER AND TITLE
MIL-P-43327B Paper, Book, Water Resistant (Text)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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